Constructability Framework Submission Form

Use this form to submit a proposed constructability entry for the reinforced concrete framework. Please provide as much detail as possible. Submissions will be reviewed and edited before being added to the framework.

Dimension *

Dropdown

Construction Methods

Country, State *

Australia, NSW

Entry name *

Off-form concrete walls near obstructions

Design question *

What specific design question does this entry answer?

When designing concrete walls near site boundaries or obstructions, what clearance is required for conventional formwork, and at what point should alternative wall systems be considered?

Design parameter *

What can the engineer measure, count, or assess from their drawings?

Available clearance between back of wall and nearest obstruction (mm)

Background

Why does this design decision matter for constructability?

Conventional off-form concrete walls require double-sided formwork with rear access for placement, bracing, and bolt-up. This assumption breaks down when walls are positioned close to site boundaries, existing fences, retaining walls, or other obstructions. It would be unwise to assume that pressing plywood against an obstruction would effectively restrain concrete, particularly at heights approaching 1000 mm or more. Where rear access is limited, the designer should consider whether conventional formwork is feasible before committing to an off-form concrete design.

Design Guidance *

Define the ranges or conditions and the rating for each. Include real evidence or incidents where possible.

Write each threshold on a new line in the format: range | rating | guidance

Ratings: Good practice, consider, concern, action required

>= 1000 mm | Good practice | Full access for formwork placement, bracing, bolt-up, and stripping. Both coil tie and threaded bar systems feasible.

700 - 1000 mm | Consider | Coil tie system feasible but access is tight — a person cannot comfortably work behind the wall. Consider whether alternatives would be simpler.

350 - 700 mm | Concern | Coil ties cannot be used (plywood + timbers + bolt = 700 mm minimum). Threaded bar only, tightened by reaching in at arm's length. Only for walls up to ~1200 mm high.

<350 mm | Action required | No conventional formwork system can be placed, braced, or bolted.

Recommended Actions

What should the engineer do if they are in the concern or action required zone?

Write actions each on new lines

- Dincel or similar permanent formwork systems that only require single-sided access. Can be rendered to achieve an off-form concrete appearance if required by the client

- Blockwork construction - may be structurally compliant depending on the design requirements (engineer to verify). Can also be rendered to match a concrete finish if aesthetics are a concern

- Reposition the wall forward to provide adequate clearance - however this is typically only viable if the client's brief allows it, as walls are often positioned near boundaries for a reason

- Precast or tilt-up panels - dependent on project scope. For larger projects with crane access this may be logistically feasible. For smaller projects without a crane, an in-situ solution is more realistic

Evidence Basis

How was this guidance established? Industry consultation, literature, project experience, incident reports?

Conventional formwork requires rear clearance for bracing and bolt-up access. Confirmed through literature + industry consultation: deriving the clearance thresholds. A concrete upturn wall designed as off-form concrete near a site boundary was found to be infeasible during construction due to zero clearance behind the wall and no access to adjacent land. The design was revised to a Dincel permanent formwork system.

References:

Dincel Structural Walling. (2026). Construction Manual. Dincel Structural Walling.
Hurd, M. K. (2005). Formwork for Concrete (7th ed.). American Concrete Institute.
Jha, K. N. (2012). Formwork for Concrete Structures. Tata McGraw Hill Education Private Limited.

Example images

Optional — add images showing examples of concern, acceptable, and recommended approaches to bring the entry to life.

off-form-detail - ...

on-site-dincel - D...

Add file

Example images captions

Optional — add captioning for images for which is a concern, acceptable, and recommended; and why

(off-form-detail.png) - | Concern |
Original design: off-form concrete upturn wall specified directly against the site boundary. No clearance behind for conventional formwork bracing or bolt-up access.

(on-site-dincel.jpg) - | Recommended |
Dincel (PVC permanent formwork) system installed to eliminate the requirement for rear bracing in a wall 1000 mm high.

Supporting images

Optional — add photos of the site outcomes or examples from literature

dincel-complete -...
formwork-bracin...
formwork-bracin...
formwork-literatu...
formwork-ties - D...
offform-stripped ...
dincel-bracing-1 -...

Add file

Supporting images captions

Optional — add captions for the supporting images to explain each in order

(formwork-bracing-1.jpg)
Conventional off-form wall formwork with timber bracing on both sides and coil ties at regular intervals

(formwork-bracing-2.jpg)
Formwork bracing detail — note the access and clearance required behind the wall for bracing and bolt-up

(offform-stripped.jpg)
Off-form concrete wall after formwork removal — coil tie holes visible at regular intervals

(dincel-complete.jpg)
Completed Dincel wall after concrete pour — no rear bracing required during construction

(formwork-ties.png)
(Hurd, 2005): Common formwork tie types — one-piece ties snap off after stripping, while internally disconnecting ties allow full removal from the concrete

(formwork-literature-example.png)
(Jha, 2012): Conventional timber wall formwork assembly — note the coil ties passing through both faces, requiring access and bracing on both sides of the wall

(dincel-bracing-1.png)
(Dincel, 2026): Dincel PVC permanent formwork wall panel — diagonal bracing connects externally, eliminating the need for rear-side access and coil ties

Submitter name

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